SIMON
BUSINESS SCHOOL
UNIVERSITY of ROCHESTER

CIS467 A3: Data Management, Data Warehousing & Data Visualization
Course Syllabus (last modified 01/06/2025)
Spring A 2025

Course Logistics:

Day and Time: **MW, 1:40PM – 3:10PM**
Location: **Schlegel Room 107**

Course Blackboard Link: https://learn.rochester.edu/ultra/courses/_102960_1/outline
Lab: Thursdays 1:40 PM to 2:40 PM (starting on week 2), Room **Schlegel Room 107**

Instructor:
Professor Mikhail Lysyakov
Assistant Professor of CIS mlysyako@simon.rochester.edu
Office: 3-314 Carol Simon Hall

Office Hours: Wednesdays (3:30 PM to 5:30 PM) in 3-314 Carol Simon Hall, and by appointment (in Zoom) outside scheduled hours

Professor's Zoom link - *NOT the same as class Zoom link*: https://rochester.zoom.us/j/7824895921

Teaching Assistants: TBA
MS student -
Zoom office hours: TBA
Zoom office hours link: TBA

PhD student
Zoom office hours: TBA
Zoom office hours link: TBA

PhD student
Zoom office hours: TBA
Zoom office hours link: TBA

Lab Instructor: Chengcheng Li cli83@ur.rochester.edu

Zoom office hours for labs: TBD
Zoom link for labs' office hours (personal, the link could be changed): TBA

Contents:

1. Course Description
2. Learning Objectives
3. Textbooks and Required Material
4. Communication
5. Assignments, Case Studies, or Projects
6. Evaluation and Grading
7. Inclusivity Statement
8. Accommodations and Accessibility
9. Course Delivery and Credit-Hour Policy Adherence
10. AI Use Policy
11. Academic Integrity
12. Professionalism and Attendance
13. Course Schedule/Roadmap

Course Information

1. Course Description

Companies and organizations in every industry rely heavily on their data, not only for the day to day tactical needs of running the business but also to generate information and analysis to support problem-solving and decision-making in a wide variety of applications. Database design and management along with systems/techniques for data retrieval, transformation and analysis have the potential to play a key role in business strategy and to create business value and competitive advantage.

CIS 467 builds an understanding of system data design to enable analytics as well as the concepts and tools necessary to leverage system data to meet an organization's analytics needs. The course covers an introduction to relational databases, MySQL programming skills for data retrieval, a conceptual framework for analytics activities surrounding data warehousing, and data visualization using Tableau.

2. *Learning Objectives*

At the end of the course, students will be able to:

- Write and execute queries in MySQL. Insert, update and delete records in MySQL.
- Design relational databases based on specifications from real world systems. Analyze database design according to the rules of data normalization. Create databases in MySQL using DDL statements. Create EER diagrams in MySQL Workbench
- Insert, update and delete data in MySQL databases
- Design data warehouses from a given relational database and/or to meet the needs of a given business application. Create ETL code to transport data from a transactional database into a data warehouse for further analytics.
- Use Tableau to create data visualizations, including dashboards, and to present data analysis using data visualization techniques. Critically evaluate the effectiveness of a variety of data visualization formats.

3. Textbooks and Required Material

Textbook (strongly recommended): Murach, Joel. Murach's MySQL. Murach: 3rd Edition.
Textbook (optional): Lemahieu, Wilfried, et al. Principles of Database Management. Cambridge University Press: 2018.

Software: MySQL 8.0 or later
MySQL Workbench
Tableau Desktop

I strongly encourage you to install the software on your local computer if possible.

There are many good references for MySQL and writing SQL queries, including many free, on-line resources. Students are encouraged to use online resources to search for relevant material and solutions to questions that come up as they complete the coursework. Other useful print references that I can recommend:

Molinaro, Anthony. SQL Cookbook. O'Reilly: 2005
Sleeper, Ryan. Practical Tableau. O'Reilly: 2018

4. Communication

Professor Lysyakov's office hours (in-person):
Wednesdays (3:30 PM to 5:30 PM)
& by appointment

Office: 3-314 Carol Simon Hall

My Zoom Office Hours Meeting link: https://rochester.zoom.us/j/7824895921

Outside of regularly scheduled office hours, email is the best way to get in touch with questions on class materials or assignments, or to setup an appointment for a Zoom meeting at a mutually convenient time.

The most recent version of the syllabus and course schedule will be posted on Blackboard. Please check Blackboard for any course announcement or schedule changes. Important announcements will also be emailed to your University email account.

5. Assignments, Case Studies, or Projects

Homework Assignments: Assignments will be posted on-line for you to download. Completed assignments should be uploaded to Blackboard on the link provided. Late assignments will receive a 10 point reduction penalty for 0-24 hours late, a 20 point reduction penalty for 24-48 hours late, and a 30 point reduction penalty for more than 48 hours late. Unless otherwise specified in the assignment instructions, assignments are submitted individually and must reflect your own original work; plagiarism will be considered a violation of academic integrity.

Quizzes: Quizzes will be taken on **Blackboard and/or** on **PAPER** during the class period. For **paper** quizzes, **PLEASE BRING A PEN OR A PENCIL**. If you are not able to take the quiz during the scheduled time, please let Professor Lysyakov know in advance to make alternative arrangements. **Paper** Quizzes are open book/notes but **NO laptops/Internet/Google/mobile phones**, and **NO** communication is permitted during the test with anyone, within the class or external. **Blackboard** Quizzes are open book/notes but **NO Internet/Google/mobile phones**, and **NO** communication is permitted during the test with anyone, within the class or external More information about the quizzes will be available shortly before the first quiz.

Final Project: The final project will be posted in the last 2 weeks of the class, and due during finals week. The final project is a **group project**. Only one project per group will be submitted. More information about the final project will be available shortly before it is posted.

6. Evaluation and Grading

The course grade is based on the following:

Homework Assignments	(3 x 10% each) 30%
Quizzes	(2 x 15% each) 30%
Professionalism* and Class Participation	*10%*
Final Project	30%
Total	100%

Team members will receive the same grade for team assignments unless peer evaluations warrant otherwise. Please review the Team Expectations section of the Simon School Student Handbook which specifies team expectations, conflict resolution, and peer reviews. All grades, as well as grade distribution and other pertinent statistics, will be put in the Blackboard Gradebook for the course.

Team Peer Evaluation: All students will have the opportunity to evaluate their peers regarding team work during the term. This evaluation can substantially affect team members grades and should be taken seriously. If you score someone below a 9, you need to provide the reason for the lower score.

7. Inclusivity Statement

At Simon, we firmly believe that a diverse, equitable, and inclusive learning environment benefits all members of our community, and we are committed to working toward lasting, systemic change in our efforts to educate the next generation of business leaders.

We strive to foster and cultivate an environment where every member of the Simon community feels supported, valued and welcome. In this classroom, students will have the opportunity to express the preferred name and pronoun by which they would like to be addressed. I (like many people) am still in the process of learning about diverse perspectives and identities. If something is said or done in class (by anyone) that makes you feel uncomfortable, please bring it to my attention privately (directly or anonymously through a staff member you trust)

8. Accommodations and Accessibility

Both Simon and the University of Rochester respect and welcome students of all backgrounds and abilities. In the event you encounter any barriers to full participation in this course due to the impact of a disability, please contact both your instructor and the Office of Disability Resources as soon as possible. The access coordinators in the Office of Disability Resources will meet with you to discuss the barriers you are experiencing and explain the eligibility process for establishing academic accommodations. You can reach the Office of Disability Resources at (585)276-5075 or at disability@rochester.edu . More information can be found at https://www.rochester.edu/college/disability/

9. Course Delivery and Credit-Hour Policy Adherence

Lectures will be held twice a week in the classroom. Lectures will also be recorded and posted on Blackboard.
Self-paced content, consisting of Panopto videos with accompanying slides and/or MySQL exercises, will be posted weekly on Friday afternoon or around that date. Students should plan on working through the self-paced content before Monday's class if possible. Self-paced content should take approximately 60 minutes to complete.
All course materials including slides, data sets, and example MySQL code will be posted on Blackboard.

This course follows the Simon credit-hour policy for 2.5-credit courses. The course meets twice weekly for 3 hours per week. In addition to these 3 hours of synchronous class sessions per week (in-person or via Zoom), students are **required**

to complete approximately one hour per week on average of asynchronous learning activities (e.g., videos and simulations). Please see the course schedule for the details related to asynchronous activities and any related assignments or assessments.

Students are also expected to supplement each hour of class or asynchronous learning activities with two hours of supplemental work (e.g., class readings, assignments, and project work).

The course also includes labs for 1 hour per week, which students are expected to attend.

Across the term and across all activities, students should expect to spend 37.5 hours per credit hour earned.

10. AI Use Policy

I may use ChatGPT and generative AI to enrich our curriculum. However, as we utilize these technologies, we must also ensure that they are used ethically and responsibly. This includes being transparent in their use, treating data with utmost privacy and care, behaving within the limits of the copyright, and maintaining an understanding that generative AI is a tool for assistance, not a replacement for human judgment and creativity.

The use of ChatGPT and generative AI promises to be enlightening and challenging as we explore new ways to enhance our teaching and learning experiences. Let's proceed with a commitment to employ these powerful tools with responsibility, care, transparency, and constant curiosity.

CIS467: AI Use Policy		
Homework and Final Project	**QUIZZES**	**DISCLOSURE POLICY**
Unless I Say No Students are encouraged to use generative AI when helpful unless specifically forbidden for particular student work.	***Never*** All work related to completing the quizzes must be your own. Generative AI may not be used.	***Citation + Description + Reflection*** Include a citation, description, and a reflection as prompted by the instructor.

Cautionary Note About Hallucination in Generative AI:

- Generative AI tools, while immensely powerful, sometimes generate outputs that are inaccurate, made up, or not reflective of reality. Students need to critically analyze AI-generated content and vet for factual errors and inconsistencies. Students remain accountable for the accuracy, integrity, and ethical use of the information in their work.

Data Privacy:

- Special care must be taken regarding data privacy. Using generative AI grants the use of the data to the AI tool. Data or documents that are copyrighted should not be uploaded into AI without permission from the copyright holder. Students remain accountable for the accuracy, integrity, and ethical use of the information in their work.

Data Bias:

- Special care must be taken regarding data bias. Some of the information provided via AI may be biased. Examples of bias include racial, gender, nationality, religion, etc. Students remain accountable for the accuracy, integrity, and ethical use of the information in their work.

11. Academic Integrity

Cheating will not be tolerated. Students should read carefully the statement on academic integrity in the student handbook. Anyone caught will be punished to the full extent allowed under the University rules.
Individual written assignments are to be completed individually. Students may collaborate on the content of the assignment but must turn in their own individual work. Any assignment that contains work that is a duplicate/copy of work from another classmate, work turned in by a student in a previous semester, model materials posted as an example or previous semester answer keys, or material from the internet will receive an automatic zero grade and an academic dishonesty report filed.

Quizzes are taken on Blackboard and/or on paper. Paper quizzes are open book/notes but **NO laptops/Internet/Google/mobile phones**, and **No** communication is permitted during the test with any other person, from within the class or externally. Blackboard quizzes are open book/notes but **NO Internet/Google/mobile phones. Students who are found to have communicated with each other or with someone from outside the class during the quizzes will receive an academic dishonesty report filing**.
Per University rules, all students will be required to sign the following pledge on each test and the final exam:

> *Academic Honesty Certification: I affirm that I will not or did not give or receive any unauthorized help on this exam, and that all work is my own*

Simon's Code of Academic Integrity (see the section Academic Integrity Policy in the Simon School Student Handbook) states: "*Every Simon student is expected to be completely honest in all academic matters. Simon students will not in any way misrepresent their academic work or attempt to advance their academic position through fraudulent or unauthorized means. No Simon student will be involved knowingly, or unknowingly yet passively within a team, with another student's violation of this standard of honest behavior.*"

In addition to refraining from obvious forms of cheating and plagiarism:

- On assignments, do not copy or paraphrase work from each other, from students who have taken the class previously, from materials of mine distributed in a previous class, or from outside sources. Any written work should be entirely your own (or your team's, as applicable).
- Exams or assignments that include an Academic Integrity/Honesty Pledge must have the pledge signed. Submissions without the pledge signed will receive a score of zero.
- Do not obtain advice, notes, solutions, or other material from students who took the class previously in ways that would give you an unfair advantage or would undermine the learning experience for you and the class (such as, notes from past case discussions). Similarly, do not use others' case analyses posted on-line.
- Use quotation marks when quoting any text directly. Changing a few words of a sentence or longer section does not make the work your own. Independently written texts rarely have even five consecutive words in common.
- Students may not audio or video record class lectures or other classroom or laboratory activities without the instructor's permission.
- Students may not publish, distribute, or sell—electronically or otherwise—any course materials that the instructor has developed in any course of instruction in the University (e.g., presentation slides, lecture aids, video or audio recordings of lectures, and exams) without the explicit permission of the instructor.

Most forms of disallowed shortcuts are easy to detect and will be referred to the school's Academic Integrity Committee. To help prevent other students from violating academic integrity, do not pass on notes or give advice on assignments to any students who are taking the course in a later term or are taking it at the same time in a different section. Please refer to the Student Handbook for any questions regarding the Code of Academic Integrity.

If a situation in your professional or personal life prevents you from finishing assigned work in a timely manner, please contact me before the deadline to discuss how to proceed. You should also contact your Ainslie OSE advisors, if appropriate. Do not violate the Academic Integrity Policy in attempt to manage a difficult situation.

12. Professionalism and Attendance

Please review the Professional Standards Policy in the Simon School Student Handbook. Students are expected to demonstrate the same professional behavior in class as they would in a business setting.

Unprofessional behavior has a negative impact on your participation grade. Specifically,

Make every effort to attend each class. Missing class may result in a lower Professionalism and Class Participation score.

We will use Qwickly to take attendance, which requires that you bring a smartphone, laptop, or tablet to each class to access Qwickly on Blackboard.

Display your printed name tent in every class.

You are expected to be in your seat and ready for class at the beginning of each class.

Should extenuating circumstances require you to leave early, please inform me before the start of class and sit near the back of the classroom so you can leave with as little disruption as possible.

At the end of the term, you will have the opportunity to provide a peer evaluation for each of your team members, which may be used to adjust their scores.

Use of laptops and cell phones (laptops allowed)

You are allowed to use laptops and tablets during lecture, but they should **only be used for coursework** related activities and not for email, social media, or other activities not directly related to the course. Cell phones must be turned off or silenced during class. No photography or recording of any kind is allowed, without express consent from me.

Course Outline and Schedule

13. Course Schedule/Roadmap

The course schedule is *tentative* and may change as we go forward during the semester. Any changes in the schedule will be announced in class and posted on-line (assignment due dates, for example).

WEEK	DATE	TOPIC	POSTED	DUE	LAB
1	**Self-paced**	Syllabus overview, Installing MySQL Server on your laptop			
	Mon., Jan. 13	Introduction to data management and MySQL, Relational databases, EER diagrams & DB schemas			
	Wed., Jan. 15	Relational databases, EER diagrams & DB schemas Data normalization, database design for analytics	**HW1 posted**		
	Wed. Jan 15	Self-paced: Creating databases and ER diagrams in MySQL			
2	**Self-paced**				
	Mon., Jan. 20	NO CLASS (MLK day)			
	Wed., Jan. 22	Data normalization, database design for analytics MySQL queries: SELECT, FROM, WHERE clauses			
	Thur., Jan. 23				**LAB1**
	Fri., Jan. 24	**Make-up class** for Monday Jan. 20 MySQL queries: SELECT, FROM, WHERE clauses, joins			
3	**Self-paced**	Not available			
	Mon., Jan. 27	MySQL queries: WHERE advanced, Summary queries			
	Tue., Jan. 28			**HW1 due 11:59 PM**	
	Wed., Jan. 29	Summary queries, GROUP BY clauses	**HW 2 posted**		
	Thu., Jan. 30	Self-paced: Case study			**NO LAB**
4	**Self-paced**				
	Mon., Feb. 3	GROUP BY clauses, Queries with subqueries, views			
	Wed, Feb. 5	Review of first 3 weeks **QUIZ 1**			

	Thur., Feb. 6				**LAB2**
5	**Self-paced**	Self-paced: Data warehouse design for analytics			
	Mon., Feb. 10	Views Data warehousing: ETL programming I		**HW 2 due 11:59 PM**	
	Tue., Feb. 11		**HW 3 posted**		
	Wed., Feb. 12	Data warehousing: ETL programming II Data warehousing examples			
	Thur., Feb. 13	Self-paced: Data warehouse case introduction			**LAB 3**
	Self-paced				
6	**Monday, Feb. 17**	Data warehousing examples Introduction to data visualization with Tableau	**Final project posted**		
	Wed., Feb. 19	Review for Quiz 2 **QUIZ 2**			
	Thur., Feb 20				**LAB4**
7	**Self-paced**	Tableau basics			
	Mon. Feb. 24	Tableau: parameters and dashboards		**HW 3 due 11:59 PM**	
	Wed., Feb. 26	Tableau: Table calculations (last day of classes)			
	Thur., Feb. 27				**LAB5**
8					
	Wed., March 5	**Final Project due 11:59 PM**		**Final Project due 11:59 PM**	